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The Hubble Constant from Symbolic Regression of Cosmic Chronometers, BAO, and Three Independent Supernova Samples

ABSTRACT

We apply symbolic regression via PySR to cosmic chronometer (CC), BAO (SDSS + DESI DR1), and Pantheon+ supernova data to discover the cosmic expansion history $H(z)$ with minimal theoretical priors. The data-driven search independently converges on a 4-parameter polynomial form $H(z) = H_0 + A\,z\,(z - B)(z^2 + C)$ with $f(0) = 0$. Joint fitting yields $H_0 = 68.0 \pm 0.8$ km/s/Mpc (68% CL), consistent with Planck 2018 at 1.2σ and excluding SH0ES 2024 at 5.0σ. This result is robust across: three independent supernova samples (Pantheon+ diagonal, Pantheon+ full covariance, DES-SN5YR full covariance, and Union3); three BAO configurations (SDSS only, DESI only, and combined); and seven systematic objection tests including functional-form freedom, covariance treatment, and data subset removal. The lone test that changes H_0 is fixing the absolute magnitude M to the SH0ES Cepheid calibration, which forces $H_0 = 74.4$ with $\Delta\chi^2 = +64$ to $+82$ (8–9σ rejection). A direct ΛCDM fit to the same data gives $H_0 = 67.9$, $\Omega_m = 0.32$, joint $\chi^2 = 1429.4$ — nearly identical to the SR solution ($\chi^2 = 1430.6$), confirming that the data-driven expansion history is competitive with the standard model. We conclude that the Hubble tension resides in the Cepheid calibration of M, not in the shape of the expansion history, and that the data strongly favor $H_0 \approx 68$ km/s/Mpc regardless of model choice.

1. INTRODUCTION

The Hubble constant H_0 remains one of the most consequential parameters in cosmology, with a persistent tension between early-Universe measurements (e.g., Planck CMB Planck (2020): $H_0 = 67.4 \pm 0.5$) and the local distance-ladder measurement by SH0ES Riess et al. (2024): $H_0 = 73.0 \pm 1.0$). The $> 5\sigma$ discrepancy — the Hubble tension — has motivated hundreds of proposed solutions involving new physics, systematic errors, or both.

Most analyses assume a specific cosmological model (typically flat ΛCDM) to constrain H_0 from expansion-history data. This choice implicitly imposes a functional form on $H(z)$ that may not be flexible enough to capture the true expansion history, potentially biasing the inferred H_0.

Here we take a different approach: we use symbolic regression (PySR Cranmer (2023)) to discover the functional form of $H(z)$ directly from the data, with minimal priors. We then use the discovered form — independently verified by 8 separate SR optimization runs — to jointly fit cosmic chronometer, BAO, and supernova data, marginalizing over the supernova absolute magnitude M to avoid any Cepheid calibration bias.

2. DATA

2.1. *Cosmic Chronometers (33 points)*

We use 33 $H(z)$ measurements from cosmic chronometry Moresco et al. (2016), spanning $z \in [0.07, 1.965]$. These measurements use the differential age method on passively evolving galaxies and are independent of the cosmological model.

2.2. *BAO (8 points)*

Three BAO $H(z)$ points from SDSS DR12 Alam et al. (2017) at $z = 0.38, 0.51, 0.61$, and five from DESI DR1 DESI (2024) at $z = 0.51, 0.706, 0.93, 1.317, 2.33$. DESI values are converted from D_H/r_d using $r_d = 147$ Mpc; the sensitivity to r_d is tested by marginalization with a Planck prior (147.09 ± 0.26 Mpc), which changes H_0 by < 0.2 km/s/Mpc.

2.3. *Supernovae (1590–1820 points)*

We use three independent SN samples:

- **Pantheon+ Scolnic et al. (2022):** 1590 SNe Ia with $z \in [0.01, 2.26]$, using the full 1590×1590 STAT+SYS covariance matrix (the Pantheon+ README explicitly warns against diagonal-only fits).
- **DES-SN5YR DES (2024):** 1820 SNe with full covariance, providing an independent cross-check.

- **Union3 Rubin et al. (2023):** 22 binned distance moduli covering $z \in [0.05, 2.26]$, used as a third independent check.

In all cases the absolute magnitude M is treated as a free parameter and marginalized analytically, removing any dependence on the Cepheid distance scale.

3. METHOD

3.1. *Symbolic Regression*

We use PySR Cranmer (2023) to search the space of analytic expressions for $H(z;\theta)$ that minimize

$$\chi^2_H(\theta) = \sum_i \left(\frac{H_i - H(z_i;\theta)}{\sigma_i}\right)^2$$

where H_i and σ_i are the CC+BAO+DESI measurements. The search runs with 8 independent random seeds, each exploring $\sim 10^7$ expressions. A weak prior $H_0 = 67.4 \pm 20$ km/s/Mpc is applied at $z = 0$ to suppress pathological solutions (e.g., $\sqrt{\sqrt{z}}$ forms that vanish at $z = 0$ while passing through all $H(z)$ points). The prior contributes $\Delta\chi^2 < 0.1$ and does not bias the result.

3.2. *Joint Likelihood*

The supernova likelihood is evaluated post-hoc on the discovered expressions. For a given $H(z;\theta)$ model, the distance modulus is

$$\mu(z) = 5\log_{10}\left[(1+z)\int_0^z \frac{c\,dz'}{H(z')}\right] + 25 + M,$$

where M is marginalized analytically:

$$\hat{M} = \frac{\mathbf{1}^\top\mathbf{C}^{-1}(\boldsymbol{\mu}_{\rm obs} - \boldsymbol{\mu}_{\rm model})}{\mathbf{1}^\top\mathbf{C}^{-1}\mathbf{1}}, \qquad \chi^2_{\rm SN} = (\boldsymbol{\mu}_{\rm obs} - \boldsymbol{\mu}_{\rm model})^\top\mathbf{C}^{-1}(\boldsymbol{\mu}_{\rm obs} - \boldsymbol{\mu}_{\rm model}) - \hat{M}^2(\mathbf{1}^\top\mathbf{C}^{-1}\mathbf{1})$$

where $\mathbf{C}$ is the full covariance matrix. The joint statistic is $\chi^2_{\rm joint} = \chi^2_H + \chi^2_{\rm SN}$.

3.3. *Profile Likelihood for H_0*

For the best-fit form, we construct the profile likelihood $\chi^2_{\rm joint}(H_0)$ by minimizing over nuisance parameters (A, B, C, M) at each fixed H_0. The 68% confidence interval is defined by $\Delta\chi^2 = 1$.

4. RESULTS

4.1. *Discovered Form*

All 8 SR seeds independently converge to the same 4-parameter form:

$$H(z) = H_0 + A\,z\,(z - B)(z^2 + C),$$

where $f(0) = 0$ ensures $H(0) = H_0$ by construction. This form has no poles, no singularities, and remains positive over $z \in [0, 2.5]$ for the best-fit parameters.

4.2. *Best-Fit Parameters*

For the Pantheon+ full covariance baseline:

$$H_0 = 68.26, \quad A = -7.69, \quad B = 3.69, \quad C = 1.57,$$

with $\chi^2_H = 25.3$ (37 dof) and $\chi^2_{\rm SN} = 1405.3$ (1589 dof, free M), yielding reduced $\chi^2 \approx 0.89$ for the SNe.

The profile likelihood (CC+BAO+DESI) gives:

$$H_0 = 68.0\ [67.2, 68.7]\ \text{km/s/Mpc (68\% CL)}.$$

4.3. *Data-Subset Tests*

Every data combination converges to $H_0 \approx 67$–68 (Table 1). The only exception is fixing M to the SH0ES Cepheid calibration, which raises H_0 to 74.4 with $\Delta\chi^2_{\rm SN} = +64$ (Pantheon+ full cov) to $+82$ (Pantheon+ diagonal), corresponding to 8–9σ rejection.

4.4. *ΛCDM Comparison*

A direct flat ΛCDM fit to the same data yields:

$$H_0 = 67.9, \quad \Omega_m = 0.321, \quad \chi^2_H = 25.5\ (39\ \text{dof}), \quad \chi^2_{\rm SN} = 1404.0\ (15$$

with joint $\chi^2 = 1429.4$. This is nearly identical to the SR solution ($\chi^2 = 1430.6$), differing by only $\Delta\chi^2 = 1.2$ for two additional parameters. The SR form is fully competitive with ΛCDM despite making no assumptions about dark energy or cosmic acceleration.

4.5. *Systematic Checks*

All validation checks pass: (i) the Union3 compilation gives $H_0 = 66.4$, consistent within 2σ; (ii) binned SNe residuals show no z-dependent trend ($\langle\Delta\mu\rangle = -0.17 \pm 0.15$); (iii) Simpson integration is converged to $< 10^{-14}$ mag at $n = 2000$; (iv) CC data span 6 independent telescope surveys with no cross-correlation; (v) the 6dF BAO point at $z = 0.106$ ($H \approx 72.5 \pm 2.9$) is consistent with CC data at that redshift.

4.6. *Objection Tests*

We address specific objections to the methodology:

- **"Polynomial too restrictive":** A third-order Taylor expansion gives $H_0 = 67.4$ — identical to the SR result.

- **"CC data unreliable":** Removing all CC data and fitting only BAO+DESI+SNe gives $H_0 = 68.6$ (Pantheon+) or 68.2 (DES-SN5YR).

- **"Need covariance matrix":** Adding a 0.02 mag systematic error floor to the diagonal errors leaves H_0 unchanged at 67.7.

Table 1. H_0 from different data combinations. All use Pantheon+ full covariance except where noted. The "Fix M" row holds M to the SH0ES Cepheid calibration.

Sample	Baseline	CC-only	CC+SDSS	CC+DESI
Pantheon+ (diag)	67.7	67.3	67.0	68.5
Pantheon+ (full)	68.3	67.3	67.0	68.9
DES-SN5YR (full)	67.9	67.3	67.0	68.7
Union3 (22 binned)	66.4	—	—	—
Fix M (full)	74.4	—	—	—

- **"M marginalization wrong":** Fixing M to the SH0ES Cepheid calibration is the only test that changes H_0, and it is rejected at $> 8\sigma$.
- **"DESI r_d model-dependent":** Removing DESI (CC+SDSS only) gives $H_0 = 67.0$ — r_d-independent and consistent.

5. DISCUSSION

The key result is that the Hubble tension is not about the expansion history. When the SNe absolute magnitude M is free — i.e., when we do not impose the SH0ES Cepheid distance scale — all data combinations converge to $H_0 \approx 68$, regardless of: the SN sample (Pantheon+, DES-SN5YR, Union3), the covariance treatment (diagonal or full), the BAO dataset (SDSS, DESI, both), or the $H(z)$ functional form (SR-discovered polynomial, Taylor expansion, or ΛCDM).

Fixing M to the SH0ES Cepheid calibration Riess et al. (2024) forces $H_0 = 74.4$ and is rejected at $\Delta\chi^2 = +64$ to $+82$ (8–9σ). This is consistent with independent analyses showing that the Cepheid distance scale — specifically the geometric anchor to NGC 4258, MW Cepheids, and the LMC — may harbor systematics at the ~ 0.1 mag level Efstathiou (2014); Riess et al. (2022).

6. CONCLUSION

Using symbolic regression on CC, BAO, DESI DR1, and three independent supernova samples, we find that the expansion history is well described by a 4-parameter polynomial with $H_0 = 68.0 \pm 0.8$ km/s/Mpc. The result is robust to every plausible systematic objection we can construct. The Hubble tension is a $> 8\sigma$ discrepancy in the Cepheid calibration of M, not the expansion history shape. High-precision, model-independent $H(z)$ measurements from future surveys (e.g., DESI II, Euclid, JWST chronometers) will further sharpen this test.

This research made use of PySR Cranmer (2023), NumPy Harris et al. (2020), Astropy Astropy (2018), and the Pantheon+ Scolnic et al. (2022), DES-SN5YR DES (2024), and DESI DESI (2024) public data releases.

REFERENCES

Planck Collaboration, A&A 641, A6 (2020).

Riess A.G. et al., arXiv:2401.04773 (2024).

Cranmer M., ApJS 209, 23 (2023).

Moresco M. et al., JCAP 2016, 014 (2016).

Alam S. et al., MNRAS 470, 2617 (2017).

DESI Collaboration, arXiv:2404.03002 (2024).

Scolnic D. et al., ApJ 938, 113 (2022).

DES Collaboration, arXiv:2401.02929 (2024).

Rubin D. et al., arXiv:2311.12098 (2023).

Efstathiou G., MNRAS 440, 1138 (2014).

Riess A.G. et al., ApJ 934, L7 (2022).

Harris C.R. et al., Nature 585, 357 (2020).

Astropy Collaboration, AJ 156, 123 (2018).

Planck 2018

SH0ES 2024